DAY 05-ETHERNET-MAC

THEORY (10 marks)

1. What is Ethernet?
    - Widely used networking technology for wired LANs.
    - Defines how ethernet communicate over networks.
    - Used in homes, offices, schools, data centres and enterprise networks.
2. What is an Ethernet frame?
   It categorized as five
    - Destination MAC address: Who receives it?
    - Source MAC address: Who sent it?
    - Type/Length: What protocol the payload carries?
    - Payload: The actual data
    - Frame Check Sequence [FCS]: Detects error
3. What is a MAC address?
    - Network interface at Layer 2
    - Local interface Identification
4. What does **MAC** stand for: Media Access Control
5. Difference between a MAC address and an IP address?
    - MAC stands for Media Access control
    - It is a network interface at Layer 2 also known as local interface identification
    - IP Address stands Internet Protocol address
    - Works in Layer 3
    - Known for Logical Addressing and Routing.
6. What is a MAC Address Table?
   AA:AA:AA:AA:AA:AA – Fa 0/1
   BB:BB:BB:BB:BB:BB – Fa 0/2
   CC:CC:CC:CC:CC:CC – Fa 0/3
   This table drives every forwarding decision
7. How does a switch learn MAC addresses?
    - Switch learns MAC address by examining the source MAC address on the incoming frames and recording it along with ports arrived in MAC address table.
8. What is a source MAC address?
    - Identifies the Interface that sends the frame
    - Switch investigate it to learn which device is on which port.
9. What is a destination MAC address?
    - Identifies the intended Receiver.
    - A switch examine and decide whether to forward frame or not.
10. What is an unknown unicast frame?
    - Also known as flooding
    - If Destination MAC cannot find in table and after switch can't pick a port it starts flooding the frame to multiple ports.

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11. What is flooding?
    - Also known as unknown unicast frame
    - If Destination MAC not found in table, switch sends to every frame and spam it.
12. What is known unicast forwarding?
    - Also known as Forwarding
    - If Destination MAC found in table, switch sends it to the appropriate assigned port
13. At which OSI layer does Ethernet switching primarily operate: Data Link
14. Why does a switch maintain a MAC Address Table?
    - It maintains MAC address table to map MAC address to its specific ports, making it follow frames to ports on correct destination.
15. Why is Ethernet knowledge important in cybersecurity?
    - It helps cybersecurity professional understand network traffic, identify attacks such as MAC flooding/ Arp spoofing, and detect suspicious activities

PRACTICAL IDENTIFICATION (4 marks)

Record: Network adapter name, MAC address, IPv4 address, default gateway.

Network Adapter Name: Wireless LAN adapter Wi-Fi

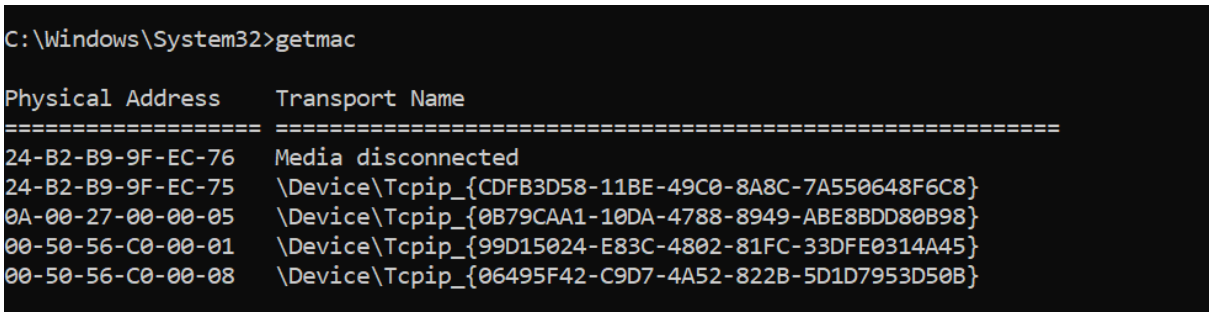
MAC address: 24-B2-B9-9F-EC-75

IPv4 Address: 192.168.100.8

Default Gateway: 192.168.100.1

SCREENSHOTS (3 marks)

Attach screenshots of: getmac, ipconfig /all, arp -a



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```
C:\Windows\System32>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Kirthikaa
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-05
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1b54:823e:5810:faec%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 621412391
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 26-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 2A-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
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Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:f40:994:3cd:b087:400d:d8af:62ee(Preferred)
   Temporary IPv6 Address. . . . . . : 2001:f40:994:3cd:204f:a7e:f058:7c96(Preferred)
   Link-local IPv6 Address . . . . . : fe80::450d:aba6:dee6:4cd7%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.100.8(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 08 October 2026 11.44.02 PM
   Lease Expires . . . . . . . . . . : 11 October 2026 12.58.25 AM
   Default Gateway . . . . . . . . . : fe80::1%18
                                       192.168.100.1
   DHCP Server . . . . . . . . . . . : 192.168.100.1
   DHCPv6 IAID . . . . . . . . . . . : 153399993
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   DNS Servers . . . . . . . . . . . : fe80::1%18
                                       192.168.100.1
                                       fe80::1%18
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.150.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 09 October 2026 11.48.37 PM
   Lease Expires . . . . . . . . . . : 10 October 2026 2.34.28 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.150.254
   DHCPv6 IAID . . . . . . . . . . . : 201347158
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

```
Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.35.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 09 October 2026 11.48.37 PM
   Lease Expires . . . . . . . . . . : 10 October 2026 2.34.28 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.35.254
   DHCPv6 IAID . . . . . . . . . . . : 234901590
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   Primary WINS Server . . . . . . . : 192.168.35.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-76
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
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```
C:\Windows\System32>arp -a

Interface: 192.168.35.1 --- 0x2
  Internet Address      Physical Address      Type
  192.168.35.254        00-50-56-e5-ee-84     dynamic
  192.168.35.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x5
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.150.1 --- 0xf
  Internet Address      Physical Address      Type
  192.168.150.254       00-50-56-e8-8d-02     dynamic
  192.168.150.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.100.8 --- 0x12
  Internet Address      Physical Address      Type
  192.168.100.1         f8-de-73-3b-39-c8     dynamic
  192.168.100.5         d0-4e-99-af-e2-04     dynamic
  192.168.100.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```